Musical Instrument Recognition Using DSP and ML

Stage 1: Single / Predominant Instrument Recognition

PROJECT MEMBERS & ROLL NUMBERS

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Introduction & Problem Statement

PROBLEM FORMULATION

CORE PROBLEM STATEMENT

“Develop a DSP + CNN based system capable of recognizing the predominant musical instrument from an audio clip.”

ENGINEERING BACKGROUND & APPROACH

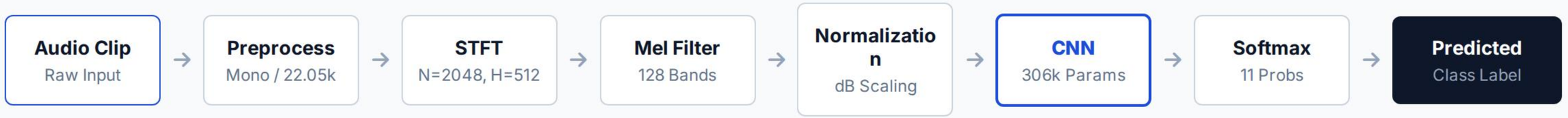
- **Audio Classification Challenge:** Identifying musical instrument timbre requires resolving intricate spectro-temporal signal characteristics.
- **Time-Frequency Transformation:** Raw audio waveforms are processed via DSP into 2D Mel spectrograms preserving acoustic harmonic patterns.
- **Deep Representation Learning:** A Convolutional Neural Network extracts translation-invariant, discriminative acoustic features.
- **Stage 1 Operational Scope:** Dedicated strictly to single/predominant instrument classification on 3-second segments.

INPUT SIGNAL	TARGET OUTPUT	INSTRUMENT SET	CLASSIFICATION TYPE
3-Second Audio Clip	1 Predicted Class	11 Target Classes	Single-Label Predominant

Overall Methodology

END-TO-END PIPELINE

HORIZONTAL SYSTEM ARCHITECTURE FLOW



1. AUDIO ACQUISITION

Duration: 3.0 seconds

Channel: Mono format

Sampling Rate: 22,050 Hz

Fixed temporal aperture

2. DSP PIPELINE

FFT Size: 2048 (Hann window)

Hop Length: 512 samples

Filter Bank: 128 Mel bands

128 × 130 Spectrogram

3. DEEP LEARNING

Blocks: 3 convolutional blocks with Batch Normalization

Pooling: MaxPool + GAP

Dense Layer: 128 units (ReLU)

Dropout 0.25 / 0.5

4. CLASSIFICATION

Output: 11-class Softmax

Decision: Argmax probability

Evaluation: Test Loss & Accuracy

Predominant instrument

Dataset and Data Split

BENCHMARK SPECIFICATIONS

ACOUSTIC BENCHMARK DATASET

IRMAS-TrainingData

Total: **6,705 audio clips** • Duration: **3.0 seconds** each • **11 instrument classes**

11 TARGET INSTRUMENT CLASSES:

- Cello [cel]
- Clarinet [cla]
- Flute [flu]
- Acoustic Guitar [gac]
- Electric Guitar [gel]
- Organ [org]
- Piano [pia]
- Saxophone [sax]
- Trumpet [tru]
- Violin [vio]
- Voice [voi]

Training Partition — 4,693 clips (70%)

Primary model optimization, feature extraction learning, and weight updates via Adam backpropagation.

Validation Partition — 1,006 clips (15%)

Unseen validation tracking, EarlyStopping checkpointing, and ReduceLROnPlateau decay triggers.

Testing Partition — 1,006 clips (15%)

Strictly held-out objective test split for final accuracy, loss, and confusion matrix evaluation.

DSP Preprocessing: Audio → Mel Spectrogram

FEATURE EXTRACTION

DETERMINISTIC DSP TRANSFORMATION CHAIN

Audio Waveform → Mono / 22,050 Hz → STFT (Hann, N=2048) → 128 Mel Filter Bank → dB Conversion → **128 × 130 × 1**

KEY PROCESSING PARAMETERS

Sampling Rate **22,050 Hz**	Window Function **Hann Window**
FFT Size **2048 samples**	Hop Length **512 samples**
Mel Filter Bands **128 bands**	CNN Input Tensor **128 × 130 × 1**

ACOUSTIC & ENGINEERING RATIONALE

“Mel spectrograms provide a compact time-frequency representation suitable for CNN-based feature learning.”

By warping linear frequencies onto the non-linear Mel auditory scale, the transformation emulates the frequency sensitivity of human hearing while preserving instrument timbral harmonics and onset dynamics.

Deeper CNN Architecture

MODEL TOPOLOGY & SUMMARY

HIERARCHICAL FEATURE EXTRACTOR BLUEPRINT

Input Tensor: 128 × 130 × 1 (Single-Channel Spectrogram)

Block 1: Conv2D (32, 3×3) + BN → Conv2D (32) + BN → MaxPool (2×2) + Drop(0.25)

Block 2: Conv2D (64, 3×3) + BN → Conv2D (64) + BN → MaxPool (2×2) + Drop(0.25)

Block 3: Conv2D (128, 3×3) + BN → Conv2D (128) + BN → MaxPool (2×2) + Drop(0.25)

Classifier: GlobalAveragePooling2D → Dense (128, ReLU) → Drop(0.5) → Dense (11, Softmax)

TRAINING CONFIGURATION & REGULARIZATION

Total Parameters: **306,155**

Optimizer: **Adam**

Batch Size: **32**

Max Epochs: **40**

Loss Function: **Sparse Categorical CE**

Callbacks: **EarlyStopping, ReduceLR**

KERAS MODEL SUMMARY SCREENSHOT

Deeper CNN

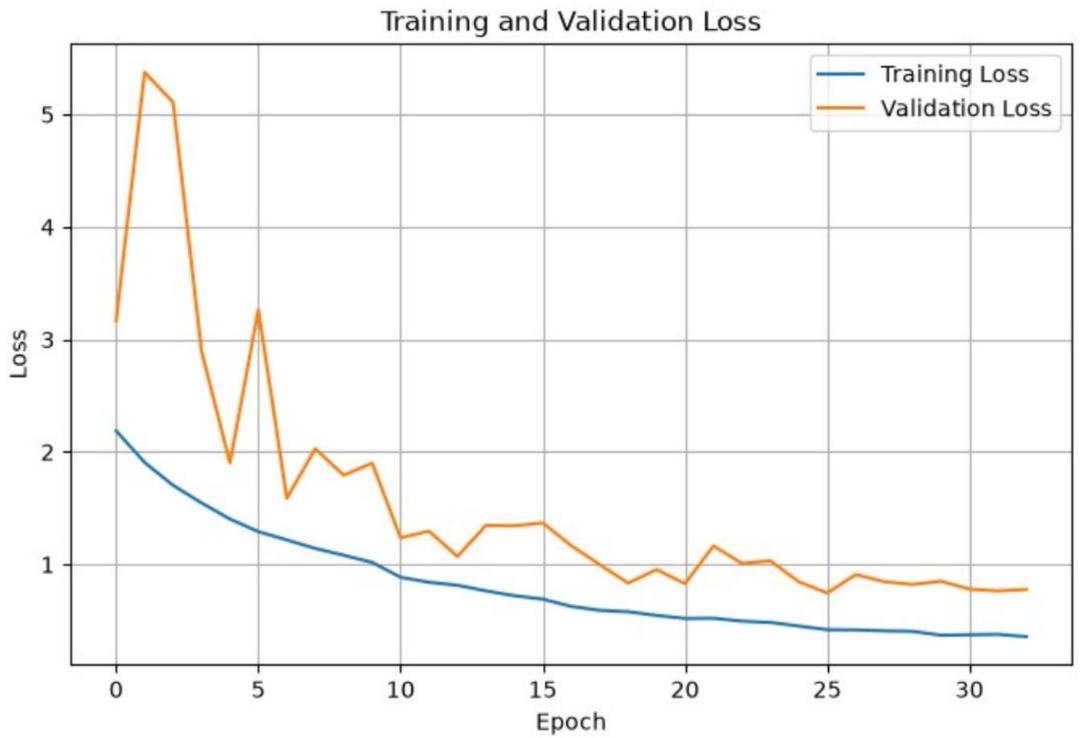
Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 128, 130, 32)	320
batch_normalization_6 (BatchNormalization)	(None, 128, 130, 32)	128
conv2d_7 (Conv2D)	(None, 128, 130, 32)	9,248
batch_normalization_7 (BatchNormalization)	(None, 128, 130, 32)	128
max_pooling2d_3 (MaxPooling2D)	(None, 64, 65, 32)	0
dropout_4 (Dropout)	(None, 64, 65, 32)	0
conv2d_8 (Conv2D)	(None, 64, 65, 64)	18,496
batch_normalization_8 (BatchNormalization)	(None, 64, 65, 64)	256
conv2d_9 (Conv2D)	(None, 64, 65, 64)	36,928
batch_normalization_9 (BatchNormalization)	(None, 64, 65, 64)	256
max_pooling2d_4 (MaxPooling2D)	(None, 32, 32, 64)	0
dropout_5 (Dropout)	(None, 32, 32, 64)	0
conv2d_10 (Conv2D)	(None, 32, 32, 128)	73,856
batch_normalization_10 (BatchNormalization)	(None, 32, 32, 128)	512
conv2d_11 (Conv2D)	(None, 32, 32, 128)	147,584
batch_normalization_11 (BatchNormalization)	(None, 32, 32, 128)	512
max_pooling2d_5 (MaxPooling2D)	(None, 16, 16, 128)	0
dropout_6 (Dropout)	(None, 16, 16, 128)	0
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 128)	0
dense_2 (Dense)	(None, 128)	16,512
dropout_7 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 11)	1,419

Training and Validation Performance

LEARNING DYNAMICS

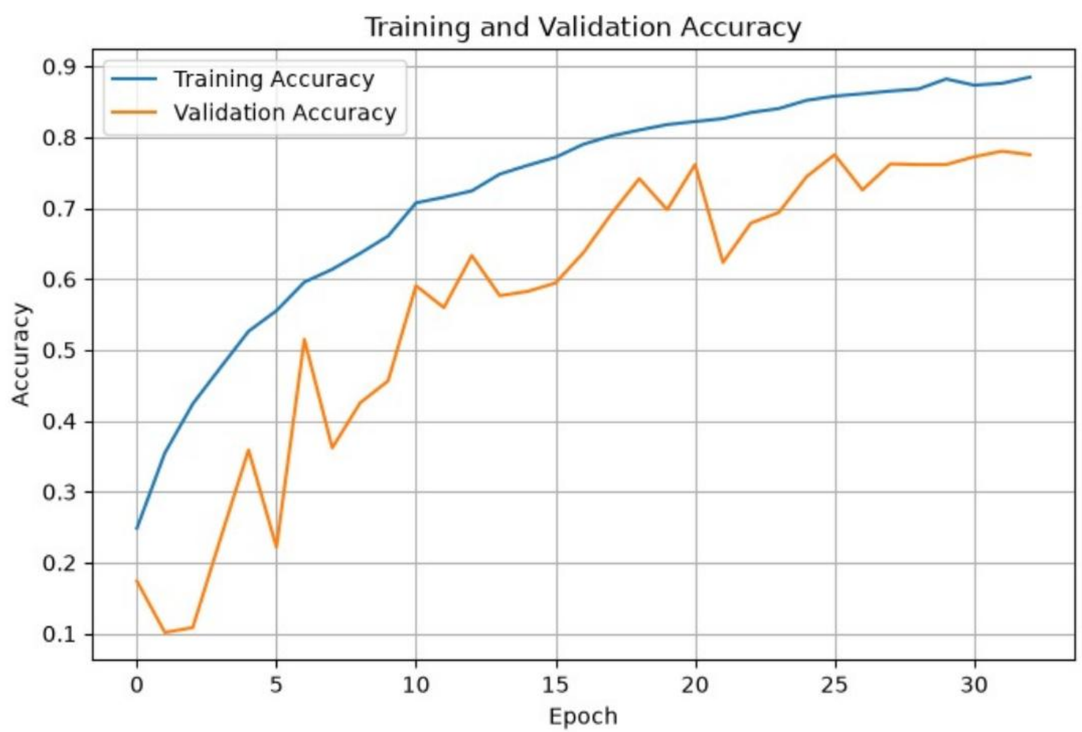
Training and Validation Loss

Cross-Entropy Loss



Training and Validation Accuracy

Accuracy Ratio [0-1]



Learning Behavior: Training loss decreases steadily, while validation performance shows fluctuations with an overall improving trend.

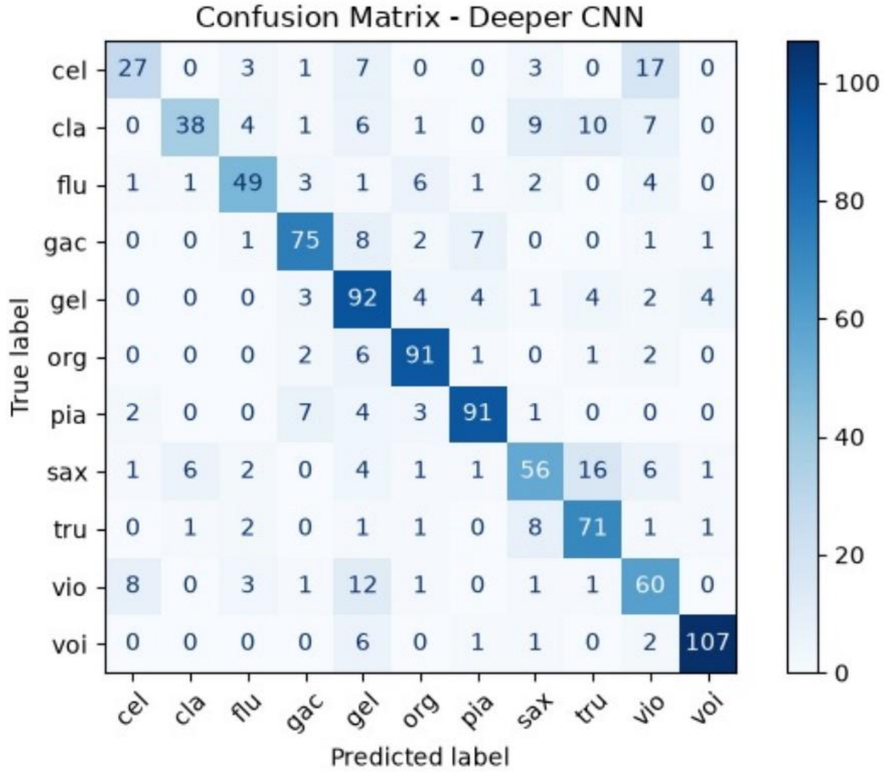
Validation Monitoring: Validation curves tracked generalization closely across epochs to detect plateau boundaries.

Training Controls: Early stopping and learning-rate reduction actively controlled overfitting and stabilized updates.

Final Results and Error Analysis

EVALUATION ON TEST SET

HELD-OUT TEST ACCURACY	HELD-OUT TEST LOSS	TOTAL TRAINABLE PARAMETERS
75.25%	0.7513	306,155



STRONG-PERFORMING CLASSES

Voice (107/116), Piano (91/108), Organ (91/102), Acoustic Guitar (75/94)

These classes show relatively strong recognition performance in the test set.

CHALLENGING CLASSES & ERROR SOURCES

Cello, Clarinet, Saxophone, Violin

Misclassifications are more common between acoustically similar instrument classes.

Comparison with Referenced Research Paper

LITERATURE BENCHMARKING

Benchmarked Reference: *"Robust CNN-based Musical Instrument Recognition with Enhanced Feature Learning"* (Sivalingam et al., 2025)

Parameter / Evaluation Metric	Our Implementation	Research Paper (Sivalingam et al., 2025)
Dataset	IRMAS	IRMAS
Audio Duration & Sampling Rate	3.0 s • 22,050 Hz	3.0 s • 22,050 Hz
STFT (FFT / Hop / Mel Bands)	2048 / 512 / 128	2048 / 512 / 128
CNN Trainable Parameters	**306,155**	≈ 440,075
Data Partitioning Strategy	**70 : 15 : 15 (Train : Val : Test)**	80 : 20 (Train : Test)
Final Evaluation Metric (Test Accuracy)	**75.25%**	≈ 78.47% (Peak Val Acc ≈ 78.37%)
Classification Output Layer	11-class Softmax	11-class Softmax

Analytical Comparison & Consistency

The implementation achieves a test accuracy in a similar range while using fewer trainable parameters. However, the comparison is not a strictly identical benchmark because the data-splitting strategies differ (70:15:15 with an isolated test set vs. 80:20). Furthermore, both studies observe identical acoustic confusions between clarinet–flute, acoustic guitar–electric guitar, and saxophone–trumpet.

Conclusion

STAGE 1 SUMMARY

- **Complete Engineering Pipeline:** A robust end-to-end DSP + CNN system was formulated, trained, and validated for predominant musical instrument recognition.
- **Acoustic Preprocessing:** STFT-based processing systematically transformed raw audio into normalized 128 × 130 Mel spectrogram tensors preserving harmonic timbre.
- **Rigorous Generalization:** The deeper CNN architecture attained **75.25% test accuracy** and a test loss of **0.7513** on the isolated IRMAS test split.
- **High Parameter Efficiency:** Operating with only **306,155 parameters**, the model showed stronger recognition performance for Voice, Organ, Piano, and Acoustic Guitar.
- **Error Analysis:** Remaining classification errors were more prominent among acoustically similar instrument classes.

“Stage 1 establishes a complete baseline and evaluation framework for musical instrument recognition.”

STAGE 2 CONTINUATION ACADEMIC PROJECT PRESENTATION • FACULTY EVALUATION

Group 4

Musical Instrument Recognition Using DSP and ML

Stage 2: Multiple Instrument Recognition (Polyphonic Mixtures)

Extension of single-instrument classification to multi-label audio mixture identification using balanced synthetic datasets, deep convolutional networks, and decision threshold optimization.

PROJECT MEMBERS & ROLL NUMBERS

Mohak Gupta (20244106) Mohit Bagri (20244107) Moksh Jain (20244108) Mudit Agarwal (20244109)

PARADIGM SHIFT

Transition from Single- to Multi-Instrument Recognition

Stage 1 vs. Stage 2 Formulation

STAGE 1 BASELINE

Single / Predominant Audio

Softmax

Target Formulation
Single-Label Ground Truth (1 of 11)

Output Layer
11-Class Softmax (Probability sum = 1.0)

Loss Function
Sparse Categorical Cross-Entropy

Inference Decision
Argmax probability selection

MULTI-LABEL EVOLUTION

STAGE 2 EXTENSION

Polyphonic Audio Mixtures

Sigmoid

Target Formulation
Multi-Hot Vector (Multiple 1s in length 11)

Output Layer
11 Independent Sigmoid Neurons [0, 1]

Loss Function
Binary Cross-Entropy (BCE) per Class

Inference Decision
Threshold-based decision (Global / Class-wise)

EXPERIMENTAL RIGOR

Dataset and Evaluation Protocol

Strict Partition Isolation

IRMAS TrainingData

6,705 Audio Clips (3.0s Mono)

Sole source for model feature learning, synthetic mixture generation, and validation threshold tuning. Partitioned via 70:15:15 stratified split.

TRAIN (70%)	VALIDATION (15%)	INTERNAL TEST (15%)
4,693	**1,006**	**1,006**
Mixture synthesis & CNN training	Threshold selection & early stopping	Unseen sanity benchmark

IRMAS TestingData

2,874 Full Tracks

⚠ **STRICT PROTOCOL ISOLATION**

Reserved EXCLUSIVELY for final official evaluation. Never used for model training, synthetic mixing, or threshold optimization.

Audio Characteristics: Excerpts of 5–20 seconds duration containing real-world multi-instrument arrangements.

> **"TrainingData was used for model development and synthetic mixture generation, while the official TestingData was reserved strictly for final evaluation."**

RESEARCH METHODOLOGY

Stage 2 Experimental Strategy

Systematic Two-Phase Investigation

Experiment 1: Baseline

Phase 1

Random Two-Instrument Mixtures

- **Mixture Generation:** Random pairs sampled from 4,693 training clips.
- **Limitation Identified:** Severe sampling skew across the 55 possible instrument combinations.
- **Role:** Serves as our polyphonic recognition baseline for threshold and aggregation analysis.

SOLVED BY

Experiment 2: Balanced Corpus

Phase 2

Balanced Two- and Three-Instrument Mixtures

- **Pair Balancing:** Exactly 85 synthetic mixtures generated for all 55 pairs.
- **Higher Polyphony:** Exactly 20 synthetic mixtures for all 165 possible triplets.
- **Core Objective:** Overcome pair under-representation and generalize to denser polyphony.

Design Rationale: Experiment 1 generated mixtures randomly, which left lower-frequency combinations under-represented. Experiment 2 enforces uniform combinatorial coverage.

EXPERIMENT 1 ARCHITECTURE

Experiment 1: Random Two-Instrument Mixtures

Synthesis Pipeline & Partitioning

Synthetic Polyphonic Synthesis Chain

Audio Clip A (Class i) + **Audio Clip B (Class j ≠ i)**

↓

DSP Acoustic Mixing Pipeline
RMS Normalization → Random Gain [0.5, 1.0] → Signal Addition → Peak Normalization

↓

Log-Mel Spectrogram
128 × 130 × 1 Tensor

Multi-Hot Target Vector
Exactly 2 active labels $[0,1]^{11}$

DATASET PARTITIONS

- Training: **4,693** two-instrument mixtures
- Validation: **1,006** two-instrument mixtures

COMBINATORIAL PAIR SPACE

11 Classes → C(11, 2) = 55 unique combinations

Each synthetic sample pairs two distinct instrument stems.

⚠ IDENTIFIED STATISTICAL SKEW

Uniform random pairing does **not** guarantee equal representation across all 55 pairs, resulting in significant empirical imbalance.

EMPIRICAL EVIDENCE

Experiment 1: Pair Distribution & Baseline

Skew Analysis & Test Benchmark

Empirical Sampling Imbalance (Examples)

Observed frequencies in the 4,693 training mixtures reveal stark disparity across combinations:

HIGHER-FREQUENCY PAIRS	
Piano + Voice	149
Organ + Voice	136
Elec Guitar + Sax	135
Elec Guitar + Piano	134
Ac Guitar + Elec Guitar	120

LOWER-FREQUENCY PAIRS	
Cello + Flute	41
Cello + Trumpet	44
Cello + Ac Guitar	44

3.6× Frequency Ratio: Distorts classifier discriminability.

EXPERIMENT 1 BASELINE

Official IRMAS Test Performance

Evaluated at Global Decision Threshold = **0.25**

MICRO PRECISION	MICRO RECALL
67.43%	57.51%

MICRO-F1 (PRIMARY)	MACRO-F1
62.08%	53.17%

Experiment 2: Balanced Two-Instrument Mixtures

Uniform Combinatorial Coverage

Combinatorial Pair Balancing

Instrument Classes	**N = 11**
Combinations of 2 Classes	**C(11, 2) = 55 unique pairs**
Mixtures Allocated per Pair	**Exactly 85 samples**
Total 2-Instrument Mixtures	**55 × 85 = 4,675**

Reproducibility Control: Synthetic mixture generator initialized with fixed **Random Seed = 42**.

Uniform Distribution Guarantee

Zero Skew

MINIMUM COUNT

85

Across every pair

MAXIMUM COUNT

85

Across every pair

Impact: Eliminates acoustic bias toward over-represented pairs (e.g., Piano+Voice) and guarantees equal acoustic training exposure for rare combinations (e.g., Cello+Flute).

Experiment 2: Balanced Three-Instrument Mixtures

Triplet Combinatorial Design

Instrument Classes	**N = 11**
Unique Triplet Combinations	**C(11, 3) = 165 triplets**
Training Mixtures per Triplet	**Exactly 20 samples**
Total 3-Instrument Train Samples	**165 × 20 = 3,300**

Each mixture superimposes 3 distinct acoustic stems with independent random gain scaling [0.5, 1.0].

Experiment 2 Validation Set Structure

Rigorous dual-polyphony validation corpus constructed without data leakage:

Two-Instrument Val Mixtures 55 pairs × 18 mixtures each	**990**
Three-Instrument Val Mixtures 165 triplets × 6 mixtures each	**990**
Total Experiment 2 Validation Set	**1,980 samples**

CORPUS ARCHITECTURE COMPARISON

Experiment 1 vs Experiment 2: Dataset Design

Structural & Volumetric Comparison

Design Parameter	Experiment 1 (Baseline)	Experiment 2 (Balanced Corpus)
Mixture Polyphony Types	Random two-instrument mixtures	**Balanced two- and three-instrument**
Two-Instrument Samples	4,693 training mixtures	4,675 balanced mixtures
Three-Instrument Samples	Not included	**3,300 balanced mixtures**
Pair Distribution	Unequal (Range: 41 to 149)	**Exactly 85 per pair (Zero skew)**
Unique Pairs Covered	55 possible (unbalanced)	55 balanced
Unique Triplets Covered	Not included	**165 possible (balanced @ 20 each)**

EXPERIMENT 2 FINAL TRAINING COMPOSITION

4,693 (Single) **+ 4,675** (2-Inst) **+ 3,300** (3-Inst) **= 12,668 Samples**

VALIDATION CORPUS

1,980 Mixture Samples

Mixture Generation & Multi-Label CNN

1. DSP Acoustic Pipeline & Network Topology

MIXING PIPELINE DETAILS

22.05 kHz Mono → 3.0s → RMS Norm → Random Gain (0.5–1.0) → Sum → Peak Norm → 128-band Mel Spectrogram → dB Scale

CNN TOPOLOGY (INPUT: 128 × 130 × 1)

- **Block 1:** Conv2D(32) + BN → Conv2D(32) + BN → MaxPool(2×2) + Dropout(0.25)
- **Block 2:** Conv2D(64) + BN → Conv2D(64) + BN → MaxPool(2×2) + Dropout(0.25)
- **Block 3:** Conv2D(128) + BN → Conv2D(128) + BN → MaxPool(2×2) + Dropout(0.25)
- **Classifier:** GlobalAveragePooling2D → Dense(128, ReLU) + Drop(0.5) → **Dense(11, Sigmoid)**

2. Training Hyperparameters & Metrics

Optimizer & Loss	**Adam | Binary Cross-Entropy**
Batch Size & Epochs	**Batch 32 | Max 50 (EarlyStop)**
Best Checkpoint Epoch	**Epoch 12**

Train AUC	Val AUC
0.9095	**0.8343**

Train Binary Acc	Val Binary Acc
90.04%	**83.31%**

Validation Loss: **0.4106**

DECISION LOGIC

Global Threshold Optimization

Validation Sweep & Micro-F1 Criterion

Sigmoid Decision Boundary

Sigmoid produces 11 independent probabilities $p_k \in [0, 1]$. A threshold τ maps probabilities into binary activations: $\hat{y}_k = 1$ if $p_k \geq \tau$ else 0.

Threshold (τ)	Micro-F1 (%)	Macro-F1 (%)	Status
0.20	58.56%	59.80%	Suboptimal Micro
0.25 (Selected)	**58.94%**	**59.65%**	**Optimal Peak**
0.30	58.49%	58.87%	Declining

Criterion Rule: $\tau = 0.25$ was strictly selected because the formal optimization criterion was **maximum Micro-F1** (58.94%), not Macro-F1.

Validation F1 vs. Decision Threshold



Lower thresholds increase false positives; higher thresholds severely degrade recall on weaker instruments.

Internal Test & Long-Audio Inference

Validation Sanity & Temporal Striding

Internal Test Benchmark

Evaluated on **1,006 unseen clips** partitioned directly from IRMAS TrainingData at global threshold **τ = 0.25**.

MICRO-F1	MACRO-F1
72.19%	**72.78%**

Confirms strong generalization on identically distributed 3.0s excerpts prior to official full-track evaluation.

Long-Audio Inference Engine

IRMAS TestingData files range from 5 to 20 seconds. An overlapping sliding window decomposes full tracks into fixed apertures:

1. Temporal Segmentation	Window = 3.0s | Hop = 1.5s (50% Overlap)
2. Window DSP & Inference	128-band Mel Spectrogram → CNN → 11 Sigmoids
3. Temporal Probability Aggregation	Mean Probability across all W windows
4. Multi-Label Decision	Thresholding against optimal decision boundary

EMPIRICAL CASE STUDY

Long-Audio Example & Aggregation Analysis

Real Audio Inference & Pooling Comparison

Case Study: Full Track Inference

IRMAS Test Track

AUDIO FILE

“02. School Boy-9.wav”

Duration: **20.0 seconds** | Sliding Windows: **13 windows**

GROUND TRUTH ANNOTATION

Piano [pia] + Trumpet [tru]

MODEL PREDICTION (τ = 0.25)

- **Trumpet: 65.70%** (Detected ✓)
- **Acoustic Guitar: 26.67%** (False Positive)

Note: Piano energy did not cross the global 0.25 threshold.

Temporal Aggregation Strategy Comparison

Evaluated across all 2,874 official test tracks at global threshold **τ = 0.25**:

Method	Precision	Recall	Micro-F1	Macro-F1
Mean (Selected)	**67.43%**	57.51%	**62.08%**	**53.17%**
Max	42.04%	73.68%	53.54%	46.90%
Top-3 Mean	48.56%	69.45%	57.16%	49.73%

Engineering Takeaway: Mean probability aggregation was retained because it produced the highest Micro-F1 (62.08%) by filtering out momentary transient noise spikes.

OPTIMIZATION ON VALIDATION DATA

Class-Specific Threshold Optimization

Compensating Instrument Probability Asymmetry

Optimized Thresholds per Instrument (Validation Split)

Cello	0.175	Clarinet	0.350	Flute	0.400
Acoustic Gtr	0.175	Electric Gtr	0.475	Organ	0.450
Piano	0.125	Saxophone	0.225	Trumpet	0.200
Violin	0.175	Voice	0.125		

Range [0.125 – 0.475]: Naturally reflects varying energy and timbral dominance across instruments.

Validation Set Performance Gain

GLOBAL THRESHOLD (0.25)
Micro: 58.94% | Macro: 59.65%
Baseline

CLASS-SPECIFIC THRESHOLDS
Micro: 61.78% | Macro: 62.00%
+2.84 pp

Protocol Invariant: Optimized thresholds were completely frozen *before* running evaluation on official IRMAS TestingData.

FINAL OFFICIAL EVALUATION

Final Benchmark: Experiment 1 vs Experiment 2

Official IRMAS TestingData (2,874 Tracks)

Evaluation Metric	Exp 1 (Random)	Exp 2 (Balanced)
Micro Precision	67.43%	**70.82%**
Micro Recall	57.51%	56.76%
Micro-F1 (Primary)	**62.08%**	**63.02%**
Macro-F1	53.17%	**53.44%**

QUANTIFIED GAINS (EXP 2 VS EXP 1)

Micro-F1 Gain: **+0.94 percentage points**

Precision Gain: **+3.39 percentage points**

Macro-F1 Gain: **+0.27 percentage points**

Recall Variation: **−0.75 percentage points**

Factual Conclusion: Experiment 2 produced a higher official Micro-F1 and Macro-F1 than Experiment 1, with significantly higher precision.

OFFICIAL CLASS-WISE BREAKDOWN (EXPERIMENT 2)

Instrument	Threshold	Precision	Recall	F1 Score
Cello	0.175	0.218	0.387	**0.279**
Clarinet	0.350	0.250	0.290	**0.269**
Flute	0.400	0.431	0.558	**0.487**
Acoustic Guitar	0.175	0.606	0.507	**0.552**
Electric Guitar	0.475	0.791	0.463	**0.584**
Organ	0.450	0.577	0.291	**0.387**
Piano	0.125	0.845	0.406	**0.549**
Saxophone	0.225	0.756	0.810	**0.782**
Trumpet	0.200	0.418	0.707	**0.526**
Violin	0.175	0.570	0.521	**0.545**
Voice	**0.125**	**0.951**	**0.892**	**0.920**

PROJECT SUMMARY & SYNTHESIS

Conclusion

End-to-End Pipeline Formulation

Developed a unified DSP + ML framework using audio preprocessing, Mel-spectrogram extraction, and CNN classification.

Stage 1 Single-Instrument Baseline

Established baseline recognition using an 11-class Softmax CNN, securing **75.25% test accuracy** with **306,155 parameters**.

Stage 2 Multi-Instrument Paradigm Shift

Transitioned to polyphonic mixtures via multi-hot target vectors, 11 independent Sigmoid outputs, and binary cross-entropy.

Combinatorial Balancing (Experiment 2)

Synthesized balanced 2- and 3-instrument mixtures, mitigating empirical pair skew from Experiment 1.

OFFICIAL BENCHMARK EVALUATION

IRMAS TestingData

Stage 2 Test Performance

Evaluated on all 2,874 full-track WAV excerpts using class-specific decision thresholds.

MICRO PRECISION	MICRO RECALL
70.82%	56.76%

MICRO-F1 (PRIMARY)	MACRO-F1
63.02%	53.44%

✓ Confirmed robust generalization across diverse multi-instrument polyphony.

> **Stage 2 extends the Stage 1 baseline from single-instrument classification to multi-instrument recognition using a multi-label DSP + CNN framework.**

Thank You

Questions & Discussion



MUSICAL INSTRUMENT RECOGNITION USING DSP AND ML

Stage 2: Multiple Instrument Recognition (Polyphonic Mixtures)